

Analysis of Current Limitations and Future Improvements

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ABSTRACT

This report summarizes the main design decisions, current limitations and potential improvements. The implemented projects include a Named Entity Recognition (NER) system for mountain name extraction and a computer vision pipeline for seasonal satellite image matching using the pretrained LoFTR model.

While both solutions satisfy the required functionality, several engineering trade-offs were made to ensure simplicity, reproducibility and compatibility with limited computational resources. This document discusses the rationale behind these design choices, identifies the primary limitations of the current implementations and proposes possible directions for future improvements.

Introduction

The implemented solutions satisfy the requirements of the tasks and demonstrate the feasibility of the proposed approaches. Nevertheless, several design decisions were intentionally simplified in order to keep the implementation compact and reproducible. This report discusses the current limitations of both tasks and outlines possible improvements that could further increase robustness, scalability and overall performance.

Natural Language Processing. Named Entity Recognition

Current limitations

Limited Mountain Name Coverage in Distant Supervision

Current limitation: The lexicon used for automated BIO sequence labeling was restricted to approximately 130 unique mountain names extracted from the Wikipedia page *List of mountain peaks by prominence*. Because the dataset was annotated using a dictionary-matching approach, any mountain names absent from this inventory were erroneously labeled as outer tokens (*O*). This structural limitation introduces substantial label noise (specifically, False Negatives) into the training data, penalizing the model for contextually recognizing unlisted mountains.

Proposed improvement: The labeling inventory could be expanded by integrating comprehensive geographical databases, such as GeoNames, Wikidata or open-source topographic registries. Additionally, regular expressions or heuristic rules (e.g., matching tokens that follow indicators like "Mount", "Mt.", or "Peak") could supplement the dictionary to capture missing entities.

Expected impact: Broadening the entity coverage for dataset annotation would drastically reduce label noise, ensure clean training signals and directly improve the model's precision and recall on real-world texts.

WordPiece tokenization

Current limitation: The BERT tokenizer splits unknown or infrequent words into multiple subword tokens (for example, "Etna" may be tokenized as "E" and "##tna"). While a post-processing step during inference maps these subword predictions back to the original word level, heavy fragmentation of rare mountain names can degrade prediction confidence or lead to inconsistent label assignments across subword sequences.

Proposed improvement: Alternative transformer architectures (such as RoBERTa or DeBERTa) employing byte-level BPE tokenization strategies could be investigated to mitigate out-of-vocabulary fragmentation. Additionally, fine-tuning on a larger and more diverse dataset would enhance the model's capacity to contextually identify and correctly classify fragmented tokens.

Expected impact: Improved handling of rare mountain names will increase prediction consistency and significantly reduce tokenization-related classification errors.

Rule-Based Alias Generation

Current limitation: Alternative mountain names (aliases) were generated using manually defined heuristic rules, such as stripping the prefix *Mount*. While this approach successfully captures common naming conventions, it fails to account for irregular naming structures and can produce linguistically unnatural or highly ambiguous aliases (e.g., converting a proper name into a frequent lowercase word). Managing these edge cases currently requires maintaining labor-intensive exception lists

to prevent introducing noise into the dataset.

Proposed improvement: Future iterations could leverage external geographical knowledge bases like GeoNames, Wikidata or Wikipedia's redirect tables to automatically retrieve verified, human-validated alternative names and localized variants.

Expected impact: Transitioning to automated alias retrieval from authoritative knowledge graphs would eliminate manual rule maintenance, drastically improve pipeline scalability and ensure high-quality, unambiguous entity variants for data augmentation.

Restricted Entity Scope

Current limitation: The pipeline operates in a single-class setup, designed exclusively for recognizing individual mountain names. It does not distinguish other closely related geographical entities — such as rivers, lakes, countries or broader mountain ranges — that frequently co-occur in the same documents. This restriction can cause the model to either miss valuable geographic context or misclassify ambiguous non-mountain entities due to shared contextual features.

Proposed improvement: The annotation schema and dataset generation pipeline could be expanded into a multi-class Named Entity Recognition format, incorporating diverse fine-grained geographical categories (e.g., *LOC-Mountain*, *LOC-Range*, *LOC-Waterbody*).

Expected impact: Transitioning to a multi-class NER framework would mitigate semantic confusion between different geographic features and significantly increase the model's utility for general-purpose geospatial and topographic text processing.

Model architecture

Current limitation: The current solution relies on the standard *bert-base-cased* architecture. While BERT provides a robust and reliable baseline for token classification, it lacks recent architectural optimizations — such as disentangled attention mechanisms or dynamic masking strategies — that have enabled newer transformers to achieve superior performance in sequence labeling tasks.

Proposed improvement: Future experiments could integrate and evaluate more advanced transformer backbones, such as RoBERTa, DeBERTa or ModernBERT, leveraging the exact same training and evaluation pipeline for a direct performance comparison.

Expected impact: Transitioning to state-of-the-art architectures is expected to yield higher entity recognition precision and recall, especially when handling structurally complex sentences and highly fragmented out-of-vocabulary subwords.

Future improvements

Expansion and Diversification of the Training Corpus

Future work will focus on scaling both the volume and semantic diversity of the training dataset. This can be achieved by incorporating naturally occurring text from specialized domains, such as digital encyclopedias, travel blogs and geographical journals. Expanding the corpus will expose the model to a broader range of syntax structures and rare topographic contexts, thereby enhancing its out-of-distribution generalization.

Systematic Hyperparameter Optimization

The current training configuration was established empirically based on standard baselines. Implementing a systematic hyperparameter search — utilizing grid search or random search techniques over critical parameters such as learning rate, batch size, weight decay and learning rate schedulers — could unlock latent model capacity and mitigate potential overfitting.

Benchmarking Alternative Transformer Architectures

While *bert-base-cased* delivered satisfactory baseline performance, newer transformer backbones optimized for sequence labeling warrant investigation. A systematic benchmark evaluating architectures like RoBERTa, DeBERTa or ModernBERT under identical data split and evaluation metrics will isolate the impact of different tokenization strategies and attention mechanisms on NER accuracy.

Multilingual Entity Recognition

The current framework is restricted exclusively to English text. Extending the annotation pipeline and training procedure to support multiple languages (e.g., through multilingual models like mBERT or XLM-RoBERTa) would allow the system to handle cross-lingual topographic data and resolve localized mountain naming variants globally.

Conclusion

The developed solution demonstrates that an effective Named Entity Recognition pipeline for mountain name extraction can be built using a relatively small automatically generated dataset together with a pretrained transformer model.

Despite the limited size of the training data, the fine-tuned model achieved strong performance on the held-out test set, indicating that the proposed dataset generation strategy and training pipeline provide a solid baseline for this specialized Named Entity Recognition task.

The identified limitations mainly concern dataset coverage, tokenization artifacts and the restricted scope of the current model. The proposed future improvements focus on increasing robustness, expanding the dataset and exploring more advanced transformer architectures.

Computer Vision. Sentinel-2 image matching

Current limitations

Fixed tile size

Current limitation: The implemented pipeline processes Sentinel-2 images using fixed-size non-overlapping tiles of 1024×1024 pixels. This design choice was made to satisfy GPU memory constraints during inference while preserving the original spatial resolution of Sentinel-2 imagery. Although computationally efficient, fixed tiling introduces artificial image boundaries. Objects located near tile borders may be split across adjacent tiles, reducing the number of detected correspondences in these regions.

Proposed improvement: A possible improvement is to use overlapping tiles instead of strictly non-overlapping ones. Overlapping regions would allow image structures near tile boundaries to be processed multiple times, increasing the probability of detecting stable correspondences.

Expected impact: This modification is expected to improve matching continuity across tile borders while only moderately increasing computational cost.

Independent processing of image tiles

Current limitation: Each image tile is processed isolatedly by the LoFTR model and the individual correspondence sets are merged only as a post-processing step after the entire scene is matched. Consequently, the pipeline ignores the spatial and geometric correlations between neighboring tiles. This independent treatment introduces the risk of local homography variance, potentially reducing matching robustness and structural alignment when dealing with large, continuous geographic features that span multiple tiles.

Proposed improvement: Future versions of the pipeline could introduce spatial consistency rules to ensure that adjacent tiles agree on the geometric alignment. Alternatively, a multi-scale matching approach could be used, where the model first analyzes a downsampled version of the whole image to guide the detailed matching of individual tiles.

Expected impact: Accounting for neighboring spatial context would enforce geometric consistency across the entire scene, mitigate localized distortion errors and significantly improve the topological integrity of the final merged alignment.

Absence of geometric verification

Current limitation: The current pipeline relies exclusively on the internal LoFTR confidence scores to filter reliable correspondences. Although LoFTR produces highly accurate feature matching, geometrically inconsistent correspondences (outliers) may still survive this filtering, particularly in homogeneous areas or regions with repetitive textures, such as dense forests or agricultural fields.

Proposed improvement: A geometric verification stage based on RANSAC could be introduced after feature matching. By estimating a geometric transformation between the two images, inconsistent correspondences can be automatically removed.

Expected impact: Geometric verification would increase the reliability of the final correspondences and reduce the number of false positive matches.

No domain-specific fine-tuning

Current limitation: The solution uses the publicly available pretrained LoFTR Outdoor model without any additional training. While the Outdoor model generalizes well to Sentinel-2 imagery, it was originally trained on natural photographs rather than satellite images.

Proposed improvement: A promising direction would be to fine-tune LoFTR on a dedicated remote sensing dataset containing Sentinel-2 image pairs with ground-truth correspondences.

Expected impact: Domain adaptation would likely improve matching accuracy under challenging seasonal conditions, particularly in agricultural areas, forests and snow-covered landscapes.

Cloud contamination

Current limitation: The current implementation processes all image regions equally, including areas partially covered by clouds or cloud shadows. Since clouds often contain little useful structural information, they may decrease the quality of

detected correspondences.

Proposed improvement: Sentinel-2 quality masks or cloud detection algorithms could be incorporated during preprocessing to exclude unreliable regions before feature matching.

Expected impact: Removing cloud-contaminated regions would improve correspondence quality while reducing unnecessary computations.

Future improvements

Adaptive tile size

The current tile size was selected primarily according to the available GPU memory. A more flexible solution would automatically adjust the tile size depending on available hardware resources. Larger tiles could be used on high-memory GPUs, while smaller tiles would remain suitable for resource-constrained environments. Such adaptation would improve computational efficiency without sacrificing compatibility.

Evaluation of alternative matching models

Although LoFTR demonstrated strong performance throughout this project, several recent deep learning architectures for image matching have recently become available. Future work should systematically compare alternative architectures such as LightGlue, SuperGlue and EfficientLoFTR using the same Sentinel-2 dataset and evaluation protocol. Such a comparison would provide valuable insights into the advantages and limitations of each approach for seasonal satellite image matching.

Conclusion

The implemented pipeline demonstrates that reliable feature matching between Sentinel-2 images acquired during different seasons is feasible using a pretrained deep learning model without additional training.

Most identified limitations are related not to the LoFTR model itself but to engineering decisions made to ensure efficient inference within limited computational resources.

The proposed improvements primarily focus on increasing robustness, improving correspondence quality and extending the pipeline toward a more scalable and production-ready solution.